

The ratio in Conjecture 00000005135 has the wrong upper endpoint

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The source explicitly takes the ratio of the *structured* condition number to the *unstructured* one, and asserts an attainable upper endpoint $\sqrt{n}$. Under the standard comparison, which retains the same norm and sensitivity functional while restricting the perturbations, every defined finite positive-denominator ratio is at most one. The proposed endpoint already fails at $n = 4$.

Write $\mathcal{H}_n$ for the vector space of Hankel matrices:

$$E_{ij} = E_{k\ell} \quad \text{whenever } i + j = k + \ell.$$

Let $a(E)$ be the amplification associated with a perturbation direction. For a differentiable matrix function it is the norm of its derivative applied to E . The relevant absolute condition numbers are

$$\kappa_{\text{u}} = \sup_{\|E\| \leq 1} a(E), \quad \kappa_{\text{s}} = \sup_{\substack{\|E\| \leq 1 \\ E \in \mathcal{H}_n}} a(E).$$

Any common positive normalization for relative condition numbers cancels in their ratio. The structured domain is a subset of the unstructured domain. The defining least-upper-bound property therefore gives

$$\kappa_{\text{s}} \leq \kappa_{\text{u}}.$$

This standard structured/unstructured comparison is also explicitly stated by Arslan, Noferini, and Tisseur[1], who interpret a small ratio in this direction as a benefit of preserving structure.

Whenever $0 < \kappa_{\text{u}} < \infty$ and the ratio is defined,

$$r = \frac{\kappa_{\text{s}}}{\kappa_{\text{u}}} \leq 1.$$

Consequently the supremum over *any* family of these ratios, when it exists as a finite least upper bound, is also at most one. In dimension four the asserted upper endpoint is $\sqrt{4} = 2 > 1$, so it cannot be attained, either by an individual pair of condition numbers or by a supremum over symbol pairs. The upper-end attainment clause of the conjecture is false.

The positivity condition does not discard a possible endpoint witness: a finite ordinary ratio equal to two requires a nonzero denominator. Zero-over-zero and infinite-over-infinite do not define such a real ratio. This argument applies to the standard common-norm comparison. Comparing a parameter-vector norm on one side with a differently scaled matrix norm on the other would be a different definition; no such change is specified in the source.

Formal verification. The Lean 4.19.0 file `lean/Main.lean` uses only bundled `Std`. It defines Hankel matrices by all anti-diagonal equalities and defines both condition numbers by the full least-upper-bound predicate on their actual perturbation domains. The amplification and norm functions are arbitrary: proving the statement for all such functions covers every choice arising from a matrix derivative.

The elementary scalar laws are packaged as the universally quantified `OrderedScale` interface. They are transitivity, positive multiplication cancellation, the multiplicative unit law, and ordinary strict-order facts. They are satisfied by real scalars; an explicit integer model is also included to verify consistency of this interface. The result is not restricted to integer-valued condition numbers. The equation $r\kappa_u = \kappa_s$ represents a ratio without division. The final theorem excludes a supremum of two for *every* family of admissible Hankel ratios in dimension four, using the actual leastness property of the supremum. It is named

`conjecture_5135_endpoint_false`.

Its printed axiom list is empty. No `sorry`, `admit`, `native_decide`, or extra axiom is used.

References

- [1] B. Arslan, V. Noferini, and F. Tisseur, *The Structured Condition Number of a Differentiable Map Between Matrix Manifolds, with Applications*, MIMS EPrint 2017.36, especially the comparison on page 2 and Section 2. <https://eprints.maths.manchester.ac.uk/2579/>.